

Nilambar Pal

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LinkedIn: linkedin.com/in/nilambar-pal — GitHub: github.com/croddy — Portfolio: crod.in

Education

SRM University – AP

2025 – Present

Bachelor of Technology in Computer Science and Engineering Undergraduate

Experience

Next Tech Lab

August 2026 – Present

Associate

SRM University-AP

- Working on IoT-based systems and embedded solutions at Next Tech Lab.

Technical Skills

- Languages:** JavaScript, TypeScript, C++, Shell, HTML/CSS, Java, C
- Frontend:** React, Vite, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form
- Backend:** Node.js, Express, MongoDB, Mongoose, REST APIs, JWT Auth
- Embedded:** ESP32, Edge Impulse, Sub-GHz RF, I2S Audio, Web Serial, FreeRTOS
- Tools:** Git, GitHub, Vercel, Railway

Projects & Technical Experience

ForenSight – Digital Forensic Investigation Platform

November 2025

1st Place – National Level Hackathon, MSC 2.0 2025 – Full-Stack Web Application

- Developed full-stack forensics platform (React, TypeScript, Node.js, Express, MongoDB) for case management, evidence tracking, and investigation workflows.
- Implemented JWT auth with role-based access, automated PDF case/evidence reports, and production hardening (rate limiting, sanitization, secure CORS).

Autonomous Line-Following Robot (ESP32)

February 2026

Personal Project – Embedded Systems / Robotics

- Built autonomous ESP32 robot with IR sensor array and real-time PWM-based steering control for line tracking.
- Designed power distribution and wiring for stable continuous operation.

Forest Sentinel – LoRa-Based Environmental Monitoring System

April 2026

3rd Place – AIESEC Hackathon Amravati – Embedded Systems / IoT

- Built ESP32-S3 sensor node + ESP32 base station over LoRa 433 MHz with on-device Edge Impulse audio classification (INMP441) for chainsaw/gunshot detection.
- Streamed compact JSON telemetry (class, confidence, BMP280 temp/pressure, sound level, RSSI/SNR, duplicate filtering) to a Web Serial dashboard with live status, map, and charts.

QOI Image Codec – C to Rust Port

August 2026

Port Mortem 2026 Hackathon – C to Rust Track

- Ported QOI codec from C to Rust (all opcodes, headers, validation) with byte-identical encode and pixel-identical decode on 8 images via oracle-diff pipeline.
- Added 97 spec-tied unit tests (zero Clippy warnings) and differential fuzzing on 2,000 inputs, fixing a truncated-stream bug with regression test.

Achievements & Leadership

- 1st Place – MSC 2.0 National Hackathon 2025 (Team Quintessence)** for developing *ForenSight*, a digital forensic investigation and case-management platform.
- Volunteer – Hospitality Committee, All India Sociological Conference**
Coordinated guest reception, registration desks, and on-site logistics, contributing to the smooth execution of a national-level academic conference.
- Volunteer – Schrodinger's Cat Hackathon, Singularity Lab, SRM AP**
Coordinated crowd management, on-site operations, and participant guidance, ensuring smooth execution of a national-level hackathon.